

AKHIL DEVABHAKTHUNI

Founder @ MetricMend | Senior Data & Analytics Engineer
AI-Powered Analytics • Semantic Modeling • Cloud Data Platforms

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Summary

- Senior Data & Analytics Engineer with 9+ years of experience building scalable, enterprise-grade analytics platforms across Azure, AWS, and Microsoft Fabric.
- Expert in **Power BI** (semantic modeling, advanced DAX, incremental refresh, OLS) with a strong track record of delivering executive dashboards and governed reporting solutions that enable data-driven decision-making.
- Specializes in designing **end-to-end cloud data platforms**, including ETL/ELT pipelines, lakehouse architectures (Bronze/Silver/Gold), and dimensional data models optimized for analytics.
- Founder of **MetricMend**, an AI-powered analytics platform that translates natural language into governed metrics, insights, and actionable recommendations—bridging the gap between raw data and business decisions.
- Combines deep technical expertise with product thinking to build modern analytics systems that are scalable, reliable, and aligned with real-world business needs.

Professional Experience

Founder / Analytics Engineer

MetricMend Oct 2025 – Present

- Modeled analytics workflows inspired by modern platforms (e.g., ThoughtSpot, Sigma, conversational BI systems)
- Built an AI-powered analytics product that translates business questions into structured insights, recommendations, and visualizations
- Designed a conversational analytics experience supporting natural language queries, follow-up interactions, and query history
- Implemented a governed analytics layer ensuring consistent metric definitions across all outputs
- Developed end-to-end architecture including frontend (React/Next.js), backend APIs, and data processing workflows

- Integrated visualization capabilities (bar, line, pie) and automated insight generation to support decision-making
- Designed approval-based access workflow to control and manage user access to analytics features
- Designed a flexible **data processing architecture supporting scalable compute engines**, including evaluation and integration patterns with platforms like Databricks for large-scale transformations
- Explored **lakehouse-based execution strategies** to support AI-driven analytics workflows, aligning with modern data platforms (e.g., Spark-based processing, distributed compute)
- Deployed live product: <https://www.metricmendai.com/>

CLOUD DATA ENGINEERING PROJECT

AWS End-to-End Lakehouse Implementation | Apr 2025 – Sep 2025

- Designed and implemented a **production-style AWS lakehouse architecture** using Amazon S3 with **Bronze, Silver, and Gold layers** for scalable analytics processing
- Built **incremental data ingestion pipelines** using Python (pandas, boto3, awswrangler), enabling efficient handling of continuously arriving datasets
- Structured raw data ingestion into **append-only Bronze layer with timestamp-based partitioning**, ensuring replayability and auditability
- Developed transformation pipelines to standardize and cleanse data into **Silver layer**, applying schema enforcement and data quality checks
- Engineered **Gold-layer star schema datasets (fact & dimension tables)** optimized for analytical workloads and BI consumption
- Implemented **data partitioning and Parquet optimization strategies**, significantly improving query performance and reducing scan costs in Athena
- Configured **AWS Athena external tables and metadata layer**, enabling serverless SQL querying over S3-based datasets
- Designed **data modeling strategy aligned with BI consumption**, ensuring seamless integration with Power BI dashboards
- Integrated Athena with Power BI to deliver **interactive dashboards powered by cloud-hosted analytics data**
- Built reusable **S3 ingestion utilities and modular ETL components**, improving pipeline maintainability and extensibility
- Applied **data engineering best practices** including modular pipeline design, separation of layers, and clear data lineage
- Simulated **real-world enterprise use case (sales analytics)** including customers, products, orders, and revenue metrics

Senior Data & Analytics Engineer

Promethean World | Oct 2022 - Mar 2025

- Led enterprise modernization of analytics platform, migrating legacy **SSIS ETL** → **Azure Data Factory** → **Microsoft Fabric**, improving scalability, reliability, and maintainability
- Designed and implemented a **cloud-native data platform architecture** using Azure and Microsoft Fabric, establishing standardized ingestion, transformation, and analytics layers
- Built and managed **Bronze / Silver / Gold lakehouse architecture**, enabling scalable and structured data processing for enterprise analytics
- Developed and optimized **ADF pipelines and Fabric Data Pipelines/Dataflows Gen2** with parameterization, triggers, and robust error handling to reduce operational failures
- Architected and owned **end-to-end cloud data platform**, including ingestion, transformation, storage (OneLake), and analytics consumption layers
- Built and owned **Power BI semantic models and executive dashboards** used by Sales and Marketing leadership for decision-making
- Developed advanced **DAX measures and time intelligence logic** (YTD, rolling 12 months, YoY comparisons, pipeline trends)
- Implemented **incremental refresh and dataset optimization**, significantly improving dashboard performance and scalability
- Designed **enterprise-grade reporting solutions** with **role-based security (RLS)** and governed datasets
- Enabled **self-service analytics** by publishing certified datasets and reusable semantic layers across business teams
- Designed **star-schema data models (fact & dimension)** to support scalable reporting and semantic layer consistency
- Migrated legacy **Cognos reports** → **Power BI & Tableau**, improving reporting accuracy and user adoption
- Standardized **business KPIs across Sales and Marketing**, ensuring consistent metric definitions across reports
- Implemented **data governance practices**, including workspace management, dataset certification, and access control policies
- Mentored team members on **Azure Data Factory, Microsoft Fabric, and modern data architecture best practices**
- Partnered with cross-functional stakeholders (Sales, Marketing, Operations) to deliver **data-driven insights and reporting solutions**
- Owned **end-to-end analytics lifecycle**, from data ingestion to executive reporting and governance

Senior BI Engineer

KinderCare Education | Jan 2019 – Oct 2022

- Designed and built **enterprise-scale Power BI semantic models**, replacing legacy SSAS cubes using star-schema architecture

- Developed advanced **DAX measures and KPI frameworks**, including fiscal calendars, time intelligence (YTD, YoY, rolling periods), and business-driven metrics
- Built and maintained **Power BI datasets and Dataflows**, enabling scalable **self-service analytics** across multiple business units
- Implemented **incremental refresh strategies**, significantly improving dataset performance and reducing refresh times
- Optimized report performance using **DAX Studio, query tuning, and indexing strategies**
- Implemented **Object-Level Security (OLS)** and managed dataset-level access controls for secure enterprise reporting
- Designed and maintained **data validation and monitoring dashboards** to ensure pipeline accuracy and reporting reliability
- Established **governed reporting practices**, ensuring consistency in metric definitions and access control
- Designed and optimized **SSIS ETL pipelines**, supporting ingestion from OLTP systems into analytical warehouse environments
- Implemented **incremental data processing strategies** to improve pipeline efficiency and scalability
- Partnered with infrastructure and DevOps teams to **automate deployment pipelines** using Bamboo, Octopus, and PowerShell
- Improved reporting performance through **SQL tuning, indexing, and profiler-based optimizations**

BI Developer

KinderCare Education | Sept 2017 – Dec 2018

- Developed **SSIS ETL pipelines** to move data from OLTP systems into analytical warehouse environments
- Designed and deployed **SSRS reports** with dynamic parameters and matrix reporting features
- Built stored procedures, functions, and automation workflows to support reporting operations
- Contributed to **star-schema data model enhancements** to improve query performance and reporting consistency

TECHNICAL SKILLS

Cloud & Data Platforms:

Microsoft Fabric, Azure Data Factory, AWS (S3, Athena), Azure SQL, SQL Server

Data Engineering & ETL:

SSIS, Fabric Data Pipelines, Python (ETL scripting), Incremental Loading, Bronze/Silver/Gold Architecture

BI & Analytics:

Power BI (Semantic Models, DAX, Incremental Refresh, OLS), Tableau (LOD Expressions, Performance Optimization, Dashboard Design)

Data Modeling:

Star Schema, Snowflake Schema, Dimensional Modeling, Fact & Dimension Design

Languages:

SQL, T-SQL, DAX, Python, PowerShell

Methodologies & Tools:

Agile (Scrum), Jira, Tabular Editor, DAX Studio

AI & Advanced Analytics:

LLM Integration (OpenAI), Natural Language Query Processing, Semantic Layer Design